STAT214 Chapter 5 Exploratory Data Analysis

Logic

Data and data-driven results are usually either

- summarized (e.g., using an average/mean) and presented in small summary tables
- presented visually in the form of graphs
  The task of producing insightful data visualizations spreads throughout the entire DSLC

1 A Question-and-Answer-Based Exploratory Data Analysis Workflow

1.1 The Process of EDA

The process of **EDA** involves creating

- preliminary data visualizations
- numeric summaries that elucidate the patterns and trends in the dataset

1.2 Where to Start Explorations

We recommend that you follow a **question-and-answer EDA workflow**, which involves

- using domain knowledge to guide your explorations
- asking and answering specific questions related to the domain problem

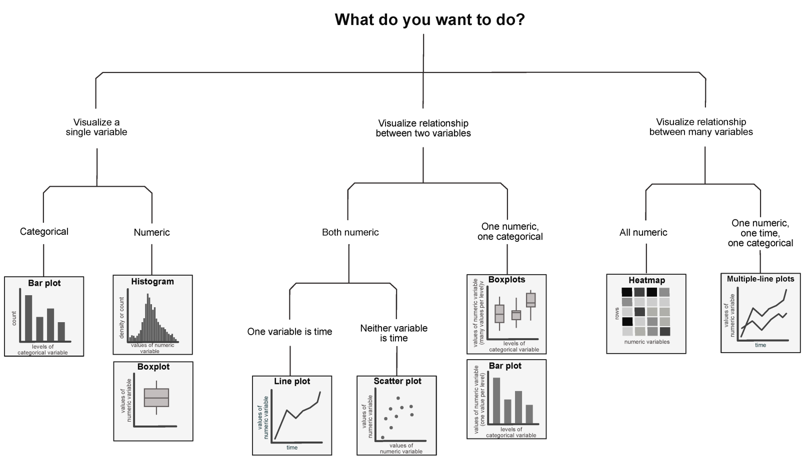
Remark: Suggestion

It is very easy to get lost in a labyrinth of exploration. Our advice is to **keep the overall project goal in mind** and use it to guide your explorations toward asking and answering questions that are relevant to the project goal.

1.3 Choosing a Data Visualization Technique

Decide which visualization technique to use:

- based on **the types of variables** that are involved in your exploratory question
- by investigating the **stability** of your data visualization judgment calls



1.4 Exploratory and Explanatory Data Analysis

Exploratory data analysis (EDA)

- preliminary data visualizations (including numeric summaries)
- some of which will contain interesting patterns and trends
- typically produced fairly quickly and roughly and are not necessarily well suited for explaining these findings to others.

Explanatory data analysis

- the process of producing polished presentation-quality figures
- communicate the particularly interesting patterns and trends to an external audience

> ⚠ **Remark: Difference Between Two Processes**
>
> The difference between the two processes lies in the time spent customizing your figures. Producing effective polished explanatory visualizations involves highlighting the elements of the preliminary exploratory plot that emphasize a clear takeaway message, and removing the elements that obfuscate it.

1.5 Tips for Explanatory Data Visualizations

Some tips for creating effective presentation-quality polished explanatory data visualizations include:

- Ensuring that the plot has a **clear takeaway message**
- Use **color** thoughtfully and sparingly
- Use **size, color, and text** to guide the audience's attention and highlight a particular element of the chart
- Use **transparency** to reduce "overplotting"
- **Titles, axes, and legend text** should be meaningful and human-readable, and legend elements should be ordered in a meaningful way.
- investigate the effect of these data visualization judgment calls in a **stability analysis**

> **Logic**
>
> Not all explorations need to be visual

2 Common Explorations

2.1 A Typical Value of a Single Variable: Mean and Median

The simplest description of a variable **combines all its values into a single digestible number** that can be thought of as a "typical" value, such as:

- mean
- median
- quantile

> ⚠ **Remark: Robustness to Outliers**
>
> The mean is much more sensitive (less robust) to outliers than the median

2.2 Histograms and Boxplots

The most common techniques for **visualizing the typical values of a single variable** are:

- histogram
- boxplot

> ⚠ **Remark**

While boxplots can certainly be used to visualize the distribution of a single variable, they are particularly helpful for **comparing** the distributions of multiple variables

2.3 The “Spread” of a Variable: Variance and Standard Deviation

- **variance**: describes how much the individual values of the variable deviate from the mean, on average
- **standard deviation (SD)**: is equal to the square-root of the variance
- **median absolute deviation (MAD)** and the **mean absolute error (MAE)**: two related measures that involve the *absolute value* difference

2.4 Linear Relationships Between Two Variables: Covariance and Correlation

Correlation and **covariance** are two related ways of measuring the strength of a **linear relationship** between two variables

2.4.1 Scatterplots

Visualize the linear relationship between two variables

2.4.2 Log Scales

Whenever you have a scatterplot in which most of the data points are bunched up in one corner of the plot, with just a few points spread out into the larger values, it can often help to represent one or both of the axes on a **log-scale**

> ⚠ **Remark**
>
> the interpretation of a linear relationship on a log-scale is one of **percentages** rather than fixed amounts

2.4.3 Correlation Heatmaps

A **correlation matrix** corresponds to a table whose entries contain the pairwise correlations between the variables of a dataset

A **heatmap** assigns a color to every number in a numeric dataset to visually convey their magnitude and overlays the colors onto the dataset.

---

3 Comparability

3.1 The Principle of Comparability

- The principle of comparability goes beyond just ensuring that the variables or summaries that you are comparing are on the same scale
- In terms of making a visual comparison, it is important to ensure that the visualization choices that you make are facilitating an apples-to-apples comparison

---

> **Logic**
>
> Before presenting any data-driven result, it is important to spend some time evaluating it using the principles of PCS

4 PCS Scrutinization of Exploratory Results

4.1 Predictability

Two ways to demonstrate the predictability:

- find external data and verify
- conduct a literature search

4.2 Stability

4.2.1 Stability to Data Perturbations

perturb the data by adding some “noise”

4.2.2 Stability to Cleaning and Preprocessing Judgment Call Perturbations

- identify any judgment calls that we made during these stages that might have influenced our results
- conduct some stability assessments to try to visualize the uncertainty arising from these judgment calls

4.2.3 Stability to Data Visualization Judgment Call Perturbations

investigate whether our findings are stable to **alternative plausible visualization judgment calls**